DIC Event App User Guide

Step-by-step workflow for event detection, review, EBSD boundary cutting, and crystallographic classification

Contents

1 Purpose Of The Software

The DIC Event App is designed to detect and analyze deformation events in digital image correlation (DIC) maps. The main workflow is:

1. Load project files such as BLN, GRX, EBSD boundary, ANG, and alignment transformation files.
2. Select a full image or crop region for event detection.
3. Tune common preprocessing steps.
4. Detect events using either mask-based detection or Hough-seed-based detection.
5. Cut events where they cross EBSD grain boundaries.
6. Analyze event shapes and send selected event types to Manual Review.
7. Manually edit, split, merge, erase, or add event pixels.
8. Classify reviewed events using crystallographic slip/twin trace vectors.
9. Save final event-level and pixel-level CSV outputs.

2 Starting The App

2.1 Using the Code Base

From the repository root, activate the conda environment and run:

```
1. git clone "https://github.com/kunwarpradip/dic.git"
2. cd path\to\dic
3. conda env create -f dic_qt\environment.yml
4. conda activate dicqt
5. python -m dic_qt.app
```

2.2 Using the Executable Files

You can directly launch the .exe file for Windows, and application file for macOS.

3 General Image Navigation

Most image panels support the same navigation controls.

Action	Behavior
Mouse wheel	Zoom in or out around the cursor.
`+` or `=`	Zoom in.
`-`	Zoom out.
`0`	Fit image to the current view.
Hold `Space` and drag	Temporarily pan the image with the hand tool.

Hover over event pixels	Highlight the event under the cursor in Manual Review.

Note: If keyboard zoom or space-drag does not respond, move the mouse over the image area first so the image panel has focus.

4 Project Files Tab

4.1 Purpose

The **Project Files** tab is the central place to define the files used by the rest of the app. Auto Detection, Boundary Cuts, Alignment analysis, and Manual Review all read from these paths.

4.2 Files

File	Required?	Use
BLN image	Required	Main DIC displacement map used for detection and display.
GRX image	Optional	Local strain map, useful for visual inspection and overlays.
EBSD aligned image	Optional	EBSD orientation image already aligned to DIC coordinates.
EBSD boundary image	Required for boundary cuts	Black/dark boundary map used to cut events at grain boundaries.
ANG file	Required for classification	Contains EBSD coordinates and Euler angles.
Transformation JSON	Required for ANG/DIC alignment	Contains transformation parameters used to map EBSD coordinates into DIC coordinates.

4.3 How To Use

1. Click **Browse** next to each file type.
2. Confirm the summary table shows expected dimensions and file status.
3. Click **Set Current Paths As Defaults** if this project should be reused often.
4. Use **Load Analysis Defaults** later to reload the saved paths.

4.4 Example

For a Ti cryo analysis, the BLN image may be a file such as:

```
Fused_BlN_step3.tif
```

The EBSD boundary image may be:

```
Aligned_EBSD_Boundary.tif
```

Warning: The app uses the file dimensions to decide whether overlays are spatially meaningful. If the BLN and boundary images do not match the same coordinate frame, boundary cuts will be wrong even if the app runs successfully.

5 Auto Detection Tab

The **Auto Detection** tab contains the main automatic event extraction workflow. Its sub-tabs are:

- **Crop**
- **Common Steps**
- **Detection**
- **Boundary Cuts**
- **Alignment**

6 Auto Detection: Crop

6.1 Purpose

The **Crop** tab selects the region used by later processing. You can process the full image or a rectangular crop. This feature lets user to first play around the smaller crop - which is faster and easier to understand the parameters required for that particular image that is being processed. Once the working parameters are figured out for the image, then it is recommended to process with the full image.

6.2 Features

Feature	**Description**
Original dimension	Shows the full BLN image size in pixels.
Use full image	Processes the complete BLN image. This is slower but produces full-field results.
Crop selector	Drag on the image to define a rectangular crop. Crop coordinates are stored in original image coordinates.
Crop X and Y	Top-left coordinate of the crop. These are usually set by dragging on the image.
Crop width and height	Crop size in pixels. These can be edited after selecting a crop.
Brightness/contrast display	Lets the user visually inspect BLN values more clearly while selecting the crop.

6.3 Example

If the original image is `10588 x 5986` and the selected crop is:

```
x = 2500, y = 1200, width = 3000, height = 3000
```

then all later detection results from that run are interpreted in global DIC image coordinates, not local crop coordinates.

7 Auto Detection: Common Steps

7.1 Purpose

Common Steps creates the processed mask used by both detection algorithms. This tab should be tuned carefully because the detection result should reflect the mask shown here.

7.2 Workflow

1. Click **Auto Process Common Steps**.
2. Inspect the image after each stage.
3. Use the stage buttons to expose only the parameters for that stage.
4. Adjust parameters until the clean/closed mask visually represents candidate event bands.

7.3 Stages

Stage	What It Does	Main Parameters
Brightness / Contrast	Maps raw BLN float values into a display/process range.	Brightness, contrast.
CLAHE	Improves local contrast without using one global contrast stretch.	CLAHE clip limit.
Threshold	Enhances ridge-like structures and thresholds candidate pixels.	Ridge sigma max, ridge percentile, Otsu multiplier.
Clean / Close	Removes small objects and bridges small gaps.	Minimum object pixels, closing radius.

7.4 View Modes

Mode	Use
Current Step	Shows the processed image for the selected step.
Raw Selected Image	Shows the selected crop/full image before preprocessing.
Overlay Step On Raw	Overlays the current processed result on the raw selected image.

7.5 Preview Resolution

- **Fast Preview** down-samples display images so sliders respond faster.
- **Full Resolution** processes the selected region at full resolution. Use this when finalizing parameters.

Note: CLAHE can be slower than other steps because it computes local contrast over many image tiles. Full-resolution CLAHE on a large BLN image can take noticeably longer than crop processing.

8 Auto Detection: Detection

The **Detection** tab supports two algorithms:

- **Mask Detection**
- **Hough Detection**

8.1 Mask Detection

8.1.1 Purpose

Mask Detection treats the final clean/closed mask from **Common Steps** as the event source. Connected mask components become events directly.

8.1.2 Algorithm

1. Use the selected crop or full image.
2. Run the same common preprocessing stages.
3. Use the clean/closed binary mask.
4. Label connected components in the mask.
5. Remove components smaller than the minimum pixel threshold.
6. Convert each remaining component into one event.

8.1.3 Output

Mask Detection creates:

- Event-level table: one row per detected event.
- Event-pixel table: one row per event pixel.
- Overlay preview: red pixels show detected mask events.

8.2 Hough Detection

8.2.1 Purpose

Hough Detection is useful when events are line-like. It first finds Hough line segments, selects seed pixels from those line segments, and then grows events from those seeds.

8.2.2 Step 1: Detect Hough Lines And Seeds

Parameter	Meaning
Hough threshold	Minimum voting strength for accepting a line segment. Higher values keep fewer, stronger lines.
Line length	Minimum accepted Hough line length in pixels.
Line gap	Maximum gap that may be bridged while forming a Hough line.
Seed spacing	Minimum spacing between selected seeds.
Use all Hough seeds	Uses all candidate seeds after spacing.
Maximum seeds	Caps seed count when **Use all Hough seeds** is off.

8.2.3 Step 2: Fill Events From Hough Seeds

Parameter	Meaning
Intensity tolerance	Allowed intensity drop while growing from a seed.
Best-fit-line tolerance	How far growing pixels may deviate from the fitted line.
Minimum intensity	Minimum blue-channel intensity accepted during growth.
Minimum points	Rejects grown events smaller than this size.
Duplicate overlap	Merges events when they share enough pixels.
Merge distance	Merges nearby compatible fragments.
Merge angle	Maximum angle difference for distance-based merging.

8.2.4 Optimization

During Hough filling, seeds inside already accepted events are skipped. This prevents many redundant seeds from growing the same event repeatedly.

9 Auto Detection: Boundary Cuts

9.1 Purpose

Boundary Cuts split detected events wherever they cross EBSD grain boundaries.

9.2 Inputs

- Detected events from Mask Detection or Hough Detection.
- EBSD boundary image from **Project Files**.

9.3 Parameters

Parameter	Meaning
Event source	Choose Mask events or Hough filled events.
Black threshold	Boundary image pixels darker than this normalized value are treated as boundary pixels.

Boundary dilation	Expands boundary pixels before cutting. Larger values cut more aggressively.
Minimum segment pixels	Discards cut fragments smaller than this many pixels.
Connectivity	Controls whether pixels are connected by edges only or by edges and corners.

9.4 Overlay Colors

Color	**Meaning**
Red	Event pixels kept after cutting.
Blue	EBSD boundary pixels.
Green	Event pixels removed by the boundary cut.

10 Auto Detection: Alignment Analysis

This sub-tab prepares crystallographic information used later by **Classification**. It is separate from the control-point alignment tab.

10.1 Crystal Setup

10.1.1 Purpose

The **Crystal Setup** step defines which crystal system and slip/twin modes are expected to be active.

10.1.2 Options

Option	**Meaning**
Crystal structure	Choose BCC, HCP, or FCC.
EBSD reference frame	Choose EDAX or Oxford convention.
Lattice parameter `a`	HCP lattice parameter used by pyramidal and twin calculations.
Lattice parameter `c`	HCP lattice parameter used by pyramidal and twin calculations.
Slip tolerance	Angle tolerance for slip modes.
Twin tolerance	Angle tolerance for twin modes.
Active modes	Checked modes are used during scoring. Their order defines preference.

10.1.3 Priority Rule

The order of active modes matters. If multiple traces are within tolerance, the app follows the selected priority order, with special handling for slip/twin conflicts according to the scoring logic.

10.2 ANG / Transform

10.2.1 Purpose

This step reads:

- `XSTEP` and `YSTEP` from the ANG header.
- Euler angles and coordinate columns from ANG core data.
- Transformation parameters from the transformation JSON file.

The app transforms EBSD/ANG coordinates into DIC coordinate space while keeping Euler angles unchanged.

10.3 Trace Vectors

10.3.1 Purpose

Trace vectors are calculated from Euler angles and the active crystal modes. These vectors represent possible slip or twin plane traces at each transformed ANG point.

10.3.2 Example

For an HCP setup with Basal and Prismatic selected:

- Basal contributes 1 trace vector.
- Prismatic contributes 3 trace vectors.
- Each ANG point has 4 candidate trace directions.

10.4 Event Shapes

10.4.1 Purpose

Event shape analysis groups boundary-cut events into categories before Manual Review.

Shape Type	**Meaning**
Linear	Event is line-like based on PCA/SVD line fit and aspect ratio.
Irregular	Event is not clearly linear or blob-like but still may be meaningful.
Blob-like	Event is dense and compact rather than line-like.
Fragmented	Event has multiple disconnected components.
Small noise	Event is below the minimum pixel threshold.

10.4.2 Send To Manual Review

The user can choose which event types are sent to Manual Review. By default, linear and irregular events are the main review candidates.

11 Manual Review Tab

11.1 Purpose

Manual Review allows the user to inspect and edit event masks after automatic detection and boundary cutting.

11.2 Event Lists

Events are grouped by type, such as:

- Linear
- Irregular
- Manual

Within each tab, events are sorted by pixel count in descending order.

11.3 Checkboxes And Selection

Action	Meaning
Check event box	Show that event in the red overlay.
Uncheck event box	Hide that event from the red overlay. The event can still be selected by hover/click.
Click event row	Select the event in the list.
Hover image event	Temporarily highlight the event under the cursor.
Locate Selected	Blink and center one selected event when it is hard to find.
Select All	Show all loaded events in the overlay.

11.4 Manual Tools

Tool	Use
Select events	Select events by clicking pixels in the image or rows in the event list.
Add pixels	Add pixels to the selected event using the brush.
Erase pixels	Erase pixels from the selected event using the brush.
Erase drawn area	Draw a freehand polygon and remove event pixels inside it. The polygon is teal/cyan.
Draw cut line	Draw a line across an event to split it into daughter events.
Grow by seed	Click a seed point to grow a new event using the manual seed-growing algorithm.

11.5 Boundary Tools

Tool	Use
Show Boundary	Overlay the EBSD boundary image from Project Files.
Cut By Boundary	Cut selected or checked events wherever they intersect boundary pixels.
Split Disconnected Pieces	If an edit leaves one event as separate islands, split those islands into separate events.

11.6 Saving And Resuming

Manual Review can be a long process, so the app supports checkpoints.

1. Click **Save Progress**.
2. The app saves edited events, event pixels, visible IDs, settings, and project file paths. The progress should be saved with extension " *.dicreview".
3. Later, click **Load Progress** to resume from the saved checkpoint.

Note: The old manual image **Choose File** button has been removed. Manual Review should be populated from auto-detection results or resumed from saved progress.

12 Classification Tab

12.1 Purpose

Classification scores the currently reviewed events against crystallographic trace vectors generated in **Auto Detection**.

12.2 Parameters

Parameter	Meaning
Max center lookup distance	Maximum allowed distance from the event center to the nearest transformed ANG point.
Minimum linearity	Minimum line-likeness required before the event can be considered likely real. Set to 0 to rely only on trace-angle matching.

12.3 Scoring Logic

For each reviewed event:

1. The event pixels are fit with a best-fit line using PCA/SVD.
2. The line direction is converted to Cartesian convention: $+x$ right and $+y$ up.
3. The event center is calculated.

4. The nearest transformed ANG point is found.
5. Slip/twin trace vectors from that ANG point are compared with the event direction.
6. If a selected trace is within its tolerance, the event is classified as **likely real**.
7. Otherwise, it is classified as **likely noise**.

12.4 Score Meaning

The score is an angle-based agreement score (1 - (Difference in angle between event and trace vector)/(Tolerance Angle)). A higher score means the event direction is closer to one of the selected crystallographic trace vectors.

12.5 Outputs

Click **Save Classified CSVs** to save:

- Event-level CSV with class, score, angle errors, center coordinates, best mode, and trace information.
- Pixel-level CSV with event pixels and classification metadata.

13 Alignment Tab: Control-Point EBSD/DIC Alignment

13.1 Purpose

The top-level **Alignment** tab is used to align an EBSD image to a DIC image using paired control points.

13.2 Workflow

1. Click **Load DIC file**.
2. Click **Load EBSD file**.
3. Use **Toggle DIC/EBSD** to switch between images.
4. Add matching control points on each image.
5. Use **Load Control Points** to paste existing point lists.
6. Click **Run Alignment** to generate the aligned EBSD image.
7. Click **Save Transformation** to save control points and polynomial coefficients.

13.3 Point Appearance

- DIC points are red with a white border.
- EBSD points are black with a white border.
- Points from the inactive image may be shown as reference points but should not be editable from the wrong image view.

13.4 Saved Transformation

The saved transformation output includes:

- Moving EBSD control points.
- Fixed DIC control points.
- Polynomial coefficients for transformed x and y.
- Fit quality information such as RMSE where available.

14 Recommended End-To-End Workflow

1. Open **Project Files** and load all available project paths.
2. Go to **Auto Detection** > **Crop** and select a crop for parameter tuning.
3. Go to **Common Steps** and tune preprocessing until the candidate mask is visually faithful.
4. Run **Mask Detection** first for band-like events, or **Hough Detection** for strongly line-like events.
5. Run **Boundary Cuts** using the EBSD boundary image.
6. In **Alignment ¿ Crystal Setup**, select crystal structure, EBSD convention, tolerances, and active modes.
7. Load ANG/transform summary and generate trace vectors.
8. Analyze event shapes.
9. Send linear and irregular events to **Manual Review**.
10. Manually edit events, split disconnected pieces, and save progress regularly.
11. Run **Classification**.
12. Save classified CSV outputs.

15 Troubleshooting

Problem	**Likely Cause And Fix**
No image appears in Auto Detection	Load a BLN image in Project Files.
Boundary cut produces zero cuts	Check that the boundary image is aligned to DIC coordinates and the black threshold is appropriate.
Boundary looks discontinuous when zoomed in	The boundary map may contain gaps or antialiasing. Try a slightly larger boundary dilation.
Mask Detection detects more components than expected	Use Full Resolution in Common Steps and confirm the final clean/closed mask.

Hough filling is slow	Too many redundant seeds may be present. Increase seed spacing, reduce max seeds, or use Mask Detection for band-like structures.
Manual Review hover cannot find events	Ensure events are loaded in Manual Review. Hidden events can still be selected, but the image and event coordinates must match.
Small selected event is hard to find	Select the event row and click **Locate Selected**.
Classification fails because ANG is missing	Load ANG and transformation JSON in Project Files, then run ANG/Transform and Trace Vectors.
Many events classified as noise	Check crystal setup, active mode priority, angle tolerance, transformed ANG coverage, and event direction convention.

16 CSV Output Overview

16.1 Event-Level CSV

The event-level CSV usually contains one row per event. Depending on the stage, it may include:

- `event_id`
- `num_pixels`
- bounding box coordinates
- event shape type
- classification result
- score
- best matching mode and trace
- event center
- best-fit-line angle

16.2 Event-Pixel CSV

The event-pixel CSV contains one row per pixel and usually includes:

- `event_id`
- `pixel_x`
- `pixel_y`
- optional source event ID
- optional shape/classification metadata

Warning: Pixel CSV files can become large for full-resolution images. Save them in project-specific output folders and keep file names descriptive.

17 Best Practices

- Tune parameters on a crop before processing the full image.
- Use Full Resolution mode before final detection if preview mode was used for speed.
- Prefer Mask Detection for finite-width band-like events.
- Use Hough Detection when events are thin and line-like.
- Use EBSD boundary cuts before manual review when grain boundaries are important.
- Save Manual Review progress often.
- Review linear and irregular events before final classification.
- Check classification overlays and saved CSVs before drawing scientific conclusions.